Device for controlling an actuator unit and system having the device, the actuator unit and a pantograph

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Abstract

This patent publication describes a device for controlling a fluid-operated actuator unit and a system comprising the device, the actuator unit, and a pantograph. The device uses multiple control valves, sensor units, and at least one computing unit to detect operating parameters and provide control signals for operating the actuator unit with fluid. The system supports controlled actuation and safe venting in pantograph applications, including fault conditions where the pantograph must be disconnected from an overhead line.

Technical summary

The invention addresses safety-relevant control of pantograph actuator systems. In railway and similar applications, a reliable connection and disconnection between a pantograph and an overhead line is essential, and the control device must be able to command both active operation and safe release in fault conditions.

The device provides a valve-based control architecture with redundant or monitored components. Sensor units detect operating parameters of the device or actuator unit, and a computing unit processes these parameters to control valves and vent or pressurize the actuator. The design supports active damping, defined operating states, and safe return or disconnection behavior.

The technical field includes fluid-operated actuators, pantograph control, pneumatic safety systems, redundant valve arrangements, sensor-based monitoring, and fail-safe venting.

Functional features highlighted for indexing

- Control of a fluid-operated actuator unit in a pantograph system.
- Multiple control valves for pressurizing, blocking, and venting actuator functions.

- Sensor units and computing unit for detecting operating parameters and issuing control signals.
- Fault-tolerant or redundant control behavior for safe pantograph retraction or disconnection.
- Application to overhead-line collector systems and actuator safety control.

Keywords

pantograph; actuator unit; fluid-operated actuator; pneumatic control; fail-safe venting; redundant valves; railway current collector; WO2023242249A1

Official source links

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- WO2023242249A1 on WIPO PatentScope
- WO2023242249A1 on Espacenet

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References

1 WIPO. WO2023242249A1, Device for controlling an actuator unit and system having the device, the actuator unit and a pantograph, published 2023-12-21.

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